

ML Library

WOC 8.0 PROJECT PROPOSAL

Prepared by Narmada P. Rangile
on December 2025



WOC 8.0

PROJECT DESCRIPTION

This project aims to build a Python **Machine learning algorithm library** from scratch. The library will support core tasks such as regression, classification, clustering, and unsupervised learning. Users will be able to import the package and easily train models on their own datasets without worrying about the low-level implementation details.

FEATURES

- **Training Logs:** During model training, the library will record key metrics(loss, accuracy, etc.) for each epoch or iteration so that the learning process is transparent and easy to monitor, also have Checkpoints in model training to recover if crash and show estimated time for training completion.
- **Hyperparameter Control:** Users will be able to configure important settings such as learning rate, number of epochs, regularization strength, degree of polynomial features, and other algorithm-specific hyperparameters.
- **Training Visualizations:** The library will include simple plotting utilities to visualize curves such as loss vs. epochs and accuracy vs. epochs, helping users diagnose underfitting or overfitting.
- **Split datasets** in train dataset, validation dataset and test dataset
- **Version control system** for models.

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TECHNOLOGY STACK

The implementation will be done in Python. For numerical computation, data handling and visualization, the following libraries will be used:

- NumPy for efficient array operations and linear algebra.
- Pandas for tabular data manipulation and preprocessing.
- Matplotlib for creating plots and training visualizations.



BRIEF IMPLEMENTATION DETAILS

The library will be built in Python using a clean, modular, object-oriented design. Each algorithm will have its own class with a common interface for training, evaluating, and making predictions. Users will be able to easily change hyperparameters, training settings, and logging options through constructor arguments or config files. Internal components like optimizers, loss functions, and metrics will be designed as plug-and-play modules, making the library easy to extend and maintain as it grows.

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TIMELINE

Week	Planned Work
Week 1	• Linear, Polynomial, and Logistic Regression implementation; • study left over content of Supervised Machine Learning course (<u>already completed 80%</u>) • study Advanced Learning Algorithms both on Coursera. • learn Exploratory Data Analysis simultaneously through all weeks
Week 2	• Implement KNN, K-Means clustering, and Decision Trees; • study "Unsupervised Learning" as needed. • Add evaluation metrics • Explore Kaggle
Week 3	• Add remaining algorithms, • Implement Neural Network from scratch: Forward , Back propagation, Weight initialization (Xavier, He), Activation functions. • Implement optimizers: (Momentum, RMSProp, Adam) • Add normalizations • start extensive testing,
Week 4	• Focused testing on real datasets, debugging, and final corrections to the library. • Documentations and Detailed report.

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ABOUT ME

Personal Details:

My name is Narmada Prasad Rangile, from Harda, Madhya Pradesh. I am pursuing B. Tech in Mining Engineering at Indian Institute of Technology (Indian School of Mines), Dhanbad.

Why should I be selected for this project?

I believe I should be selected because I genuinely love brainstorming real-world problems and solving them through programming. My passion for coding began back in 6th grade. I used to play many games, and whenever I got stuck, I would download modded APKs to experiment with them. Later, I discovered a tool that allowed me to apply custom cheat codes through scripting and that moment sparked my curiosity to understand how code truly works.

From there, I kept learning first **Python, then C, and later C++**. Each new language expanded my abilities and allowed me to build my own tools and solutions. Along the way, my interest in mathematics grew stronger, which naturally pulled me toward machine learning, especially deep learning. I was truly curious to know how AI works, and when I came across 3Blue1Brown's explanation of machine learning, I was amazed by how mathematics can lead to such powerful and intelligent applications.

I am dedicated, disciplined, and committed to completing tasks efficiently and on time. I am confident that I can devote the necessary time and effort to this project and deliver a polished, fully functional solution that meets all requirements and even enhance it further if time permits.

I am genuinely excited about this opportunity because it aligns perfectly with my passion. It offers me a chance not only to contribute meaningfully but also to learn, grow, and challenge myself with something I truly care about.